PakEmis Sector Technical Note TN05

A Hybrid Survey- and Supply-Constrained Residential Greenhouse-Gas and Air-Pollutant Inventory for Pakistan with Cross-Sector Biomass Accounting

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Abstract

Pakistan's official inventory reproduces the metered residential fossil core—its gaseous activity sits within 1.4 % of the regulator-derived quantity used here, and within 8 % on its own scope and year—but reports every residential biomass cell as not estimated. We combine regulatory statistics, two rounds of household-survey microdata and cross-sector biomass ledgers to estimate nine household fuel streams (IPCC 1A4b), with greenhouse gases and air pollutants computed from one activity basis. Firewood is triangulated by three mutually corroborating estimates, but dung is not: the agriculture nitrogen ledger implies 58.74 Mt of air-dry cake where the household survey, interpolated between its two rounds, puts 10.93 in households, a factor of 5.4 that mass conservation cannot adjudicate. The reference takes household dung from the survey (configuration S), whose two rounds themselves disagree by a factor of 1.53 per household; the configuration in which households burn all the diverted dung (L) is reported as the alternative, and the diverted dung with no observed household destination is carried explicitly as unresolved, outside the total. Across the two configurations the sector emits 24.2 Mt to 29.5 Mt CO_2e for FY2023–24 (5.4 % to 6.5 % of a ten-sector national inventory) and 283 kt to 632 kt of gross primary $PM_{2.5}$, 49 to 109 times road transport's exhaust particulate, the reference being the lower figure of each range; the dung quantity and destination are the dominant structural uncertainty in both emission layers, and the reference falls within 15 % of the independent CCAC/SEI assessment at the matched year where configuration L is 2.6 times above it. The results show that climate and primary-emission inventories can rank the same sector very differently, while intervention and exposure rankings require evidence beyond a gross-emission inventory.

Keywords: residential fuel combustion, household biomass, dung cake, black carbon, particulate matter, cross-sector closure, biomass memo convention, IPCC 2006 Guidelines, Pakistan, household air pollution

PakEmis Sector Technical Note TN05. Technical documentation of one sector module of the PakEmis release identified below, deposited with the data as part of the documentation record. The national data-description article (`paper_national/`) is the only document that states the national totals as its own result.

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Release provenance. This report documents one release of PakEmis and is generated against it. Where a sector report and the release's generated product catalogue (`outputs/pakemis_product_catalog.csv`) disagree, the catalogue is authoritative: sector reports are revised on their own cadence and may lag the products.

Release version	`1.0.0rc67`
Commit	`1876f398f85c`
Reporting period	FY2023-24 (2023-07-01 to 2024-06-30)
Baseline configuration	S (survey basis)
National total	444.96 Mt CO_2e (AR5) under S, the reference; 450.20 Mt under L
Release manifest	`13462fb4c2d1ea62` (SHA-256 of `outputs/release_manifest.json`, first 16)
Source manifest	`c1be6d314cfce1ea` (SHA-256 of `data/manifests/source_manifest.csv`, first 16)
Data-description article	`paper_national/` (ESSD, in preparation), same release

1. Introduction

Household fuel is the sector where greenhouse-gas accounting and air-quality accounting most clearly disagree about what matters.

Under the IPCC conventions, burning dung and firewood is a modest source: the CO_2 is biogenic and reported outside the national total, leaving only the methane and nitrous oxide of incomplete combustion. Under a primary-emission lens the same combustion is among the dominant sources of primary particulate in the country, emitted at breathing height inside and around dwellings by the population least able to avoid it—though translating emitted mass into exposure requires evidence this inventory does not carry (Section 10).

Both accounts are correct. They are answering different questions, and a sector that is 5.4 or 6.5 % of Pakistan's emissions (under the survey-based reference and the all-diverted dung configuration of Section 7) and, under either, one of the largest sources of its primary particulate is where the difference between those questions has the largest practical consequence.

1.1. What this paper does

We report Pakistan's residential fuel combustion for FY2023–24 from national fuel supply statistics and household-survey quantities, with an air-pollutant layer computed from the same fuel quantities, and treat the comparison between the two as the result rather than as an aside. Throughout, "layers" distinguishes the greenhouse-gas from the air-pollutant calculation, "configurations" the two dung activity bases (S, the reference, and L), and "scenarios" the structural sensitivity cases; the word inventory refers to the whole system.

The comparison is defensible here in a way it usually is not. Cross-source particulate comparisons in developing countries are typically assembled from separate studies with separate activity bases, so the ratio between two sources is partly an artefact of which datasets each author had. Every air layer in the inventory this sector belongs to is computed from that sector's own greenhouse-gas activity data—the same fuel, the same tonnage, the same population panel—so the ratio between household fuel and road transport reported in Section 8 (49 to 109 across the two dung configurations) is free of the mismatched-activity artefact, though it remains conditional on the stated factor, boundary and configuration choices.

1.2. A transfer that had nowhere to go

The sector also makes explicit a gap that had been open in this inventory and is common in others. Pakistani households burn cattle and buffalo dung, and the agriculture sector accounts for that on the supply side: it removes 28 % of cattle and 30 % of buffalo excreted nitrogen from the manure-management balance and reports it with the notation key IE—included elsewhere.

Until a residential sector existed, nowhere was elsewhere: 1.259 Mt of excreted nitrogen a year left one balance correctly and entered no other. The ledger now conserves that diverted quantity—what agriculture removes is accounted for exactly once, as household combustion or as a typed unresolved remainder—while the *destination* of the diverted mass is treated as an open question: the reference books to household combustion the consumption the household survey observes, leaving 47.8 Mt with no observed destination, outside the total; configuration L, the reference of earlier releases, books it all to household combustion (Section 4).

1.3. Contributions

1. **A fuel-anchored household energy inventory for Pakistan** across nine streams, with no household-count model and therefore nothing that can drift from the fuel actually supplied (Section 4).
2. **An executable cross-sector dung ledger** that conserves the quantity diverted from manure management while distinguishing the observed household destination from an unresolved remainder (Appendix C).
3. **Greenhouse gases and air pollutants on one activity basis**, which is what makes the source ranking in Section 8 a physical statement rather than a data-availability artefact.
4. **An explicit treatment of the biomass memo convention**, under which 65 % of household energy produces 24 % of reported emissions, and of the 65 Mt-scale biogenic CO_2 that sits outside the total.
5. **A survey-anchored firewood quantity**—the HIES 2024–25 design-weighted household measurement, corroborated by the independent 2018–19 round and by FAOSTAT's moisture-matched equivalent, replacing the flagged FAO estimate the sector previously rode—with the sensitivity still published rather than a single number (Section 7).

Method shared with the other nine sectors is documented separately (Ahmad, 2026).

2. Related Work

2.1. Household energy inventories and their activity problem

Residential fuel inventories are usually built one of two ways: from a household survey giving fuel use per household by fuel and region, or from national fuel supply attributed to the domestic sector. Surveys give richer structure and are expensive and infrequent; supply statistics are annual and coarse.

Pakistan has both in principle. The Pakistan Social and Living Standards Measurement survey carries household fuel questions, and OGRA publishes gas and petroleum supply by consuming sector. This inventory uses the supply route for the fuels that pass through a market, for the same reason the transport sector does: the quantity that a national inventory must reproduce is the fuel the country consumed, and a survey-based estimate has to be reconciled against it afterwards.

The weakness of that choice is specific and is treated in Section 10: supply statistics cover fuels that pass through a market, and the two largest household biomass fuels in Pakistan—firewood and dung—largely do not; for those two the household survey is the estimate.

2.2. Household air pollution in South Asia

Solid-fuel combustion in households is among the most studied exposure pathways in the region, and the health literature on it is far stronger than the emissions literature. What is comparatively weak is the link between the two: exposure studies measure concentrations in and around dwellings, while emission inventories estimate mass released, and translating between them requires assumptions about ventilation, stove type and time-activity patterns that neither literature settles.

This paper does not attempt that translation. It reports emitted mass on the same activity basis as the greenhouse gases, which is what an inventory can defensibly supply, and treats the exposure implication as a ranking of sources rather than a concentration estimate.

2.3. Emission factors for biomass cookstoves

The Andreae compilation (Andreae, 2019) and the Akagi compilation (Akagi et al., 2011) are the standard field-measurement syntheses for biomass combustion, and the EMEP/EEA small-combustion chapter (EMEP/EEA, 2023) is the standard source for residential fossil fuels. All carry wide ranges, because cookstove emissions depend heavily on stove design, fuel moisture and operator behaviour, and because measurement campaigns disagree.

The World Bank's 2024 Pakistan Energy Survey (World Bank, 2024), built on a nationwide 2021–22 Multi-Tier Framework survey, publishes the national stove typology this literature previously lacked: main-stove shares by locality and province, a clean-fuel main-stove share of 44.3 %, rural dominance of traditional and three-stone/open-fire stoves (71.8 %), and stove stacking at 6.2 %. It stops short of the fuel-by-device-by-end-use crosswalk a fully technology-resolved emission inventory needs, which is how this paper uses it: as the evidence base for a declared technology-mix sensitivity rather than a resolved device layer.

The biomass particulate and black carbon factors used here come from Andreae's compilation, the fossil factors from the EMEP/EEA chapter, and the crop-residue rows mirror the field-burning factors (EMEP/EEA, 2019) the agriculture sector already uses—one compilation family per burning layer, but literature values rather than Pakistani measurement throughout, which is the layer's principal limitation and is stated as such.

2.4. The biomass CO_2 convention

IPCC reports CO_2 from biomass combustion as a memo item outside national totals, on the reasoning that the carbon was removed from the atmosphere by the plant and its release is part of a short cycle, with any net loss of biomass stock accounted in the land sector instead. Methane and nitrous oxide from the same combustion are counted in full, because they are different molecules with different radiative effects and no corresponding uptake.

The convention is standard and Pakistan's own national submission states it for the same fuels (Ministry of Climate Change, Government of Pakistan, 2022). It is worth spelling out here because it produces a result that reads oddly without it: biomass is 65.4 % of household energy in this sector and 24 % of its reported emissions.

3. Study Area and Data

3.1. Nine streams, four provenances

The nine fuel streams divide into four provenance classes, and the class determines how much of this paper each stream needs. The metered fossil streams (natural gas, LPG, kerosene) are read from regulator and industry sales tables and are essentially settled; the survey-anchored streams (firewood, dung cake, coal, charcoal) take their quantities from design-weighted HIES microdata; the ledger-derived stream (crop residues, and dung cake under configuration L) is computed from the agriculture sector's mass balances rather than observed; and the expenditure-derived generator stream is priced out of the survey's fuel spending. Table 1 gives each stream's energy contribution and source; the rest of this section treats the streams in declining order of the scrutiny their provenance demands.

Stream	TJ	%	Source
Dung cake	126 827	13.6	HIES rounds, interpolated
Firewood	360 873	38.7	HIES 2024–25 quantities
Natural gas	284 553	30.5	OGRA domestic sales
Crop residues	122 192	13.1	Agriculture residue ledger
LPG	33 948	3.6	OGRA domestic sales
Kerosene	1 079	0.1	OCAC domestic sectoral sales
Generators	2 495	0.3	HIES expenditure / petrol price
Coal	341	<0.1	HIES 2024–25 quantities
Charcoal	246	<0.1	HIES 2024–25 quantities
Total	**932 554**	**100.0**	

Table 1: Household fuel input, FY2023–24, reference. Four provenances—a cross-sector ledger, a household survey (Pakistan Bureau of Statistics, 2025, 2019), a metering regulator (Oil and Gas Regulatory Authority, 2024; Oil Companies Advisory Council, 2024), and an expenditure-derived stream priced at the administered petrol rate (Government of Pakistan, 2025)—differing in quality by an order of magnitude; Section 10 treats them separately rather than reporting a single activity uncertainty.

3.2. The fossil streams

Natural gas, LPG and kerosene are metered, billed and reported by the regulator with a domestic sector breakdown. They are the best-constrained activity data in the sector and carry correspondingly narrow uncertainties: 10.8 % on gas CO_2, 20.9 % on LPG, 20.2 % on kerosene.

Natural gas alone is 15.96 Mt of CO_2—66 % of the sector's reported emissions from 31 % of its energy—because it is the only major stream whose CO_2 is reportable rather than a memo item.

3.3. Dung cake: one diverted quantity, two destination readings

The quantity agriculture diverts as dung fuel is derived from that sector's own nitrogen flow: the same herd, the same species categories, the same burned-fuel shares that drive agriculture's

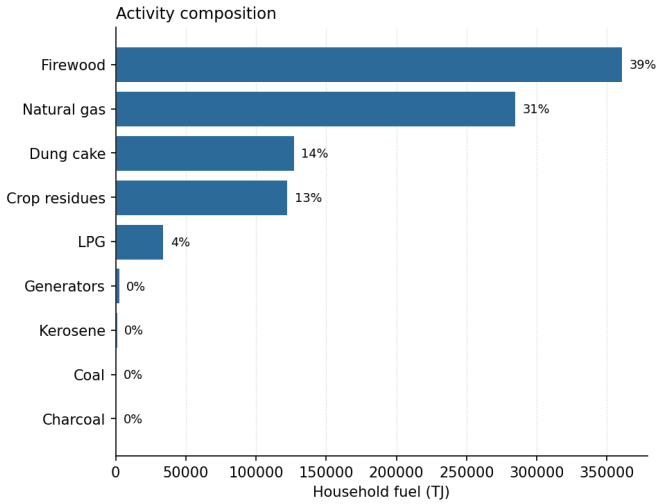


Figure 1: Activity composition. Biomass—dung, firewood, crop residues and charcoal together—is 65.4 % of household energy, which is the fact behind both the particulate result and the sector's uncertainty.

removal, converted from nitrogen to volatile solids at IPCC's rates for the same animal categories and region (Appendix C). The chain runs 38.77 Mt of ash-free volatile solids → 51.69 of total dry solids (ash restored) → 58.74 Mt of air-dry cake (moisture restored, the mass the as-received calorific value pairs with; fractions from proximate-analysis literature (Szymajda and Łaska, 2019); Table 4).

What that diverted mass does next is read two ways. The reference books to household combustion the cake households report burning in HIES: the two rounds (6.50 Mt in 2018–19, 11.97 in 2024–25) anchor a per-household quantity, interpolated between the rounds and held at the first round's rate before it, and FY2023–24, which lies between them, takes 10.93 Mt (9.62 of total dry solids). The remainder's destination is left unresolved, outside the total. Configuration L books all 58.74 Mt to household combustion: a destination assumption for the diverted dung, the reference of earlier releases, and not a reading of household use. Section 7 carries both.

The survey leaves its own questions open. The two rounds disagree by a factor of 1.53 per household. The design SE of the dung total is 6.6 % in 2024–25 and 4.1 in 2018–19, and in 2024–25 the top 5% of reporting households hold 44.7 % of the reported dung, against 19.6 in the earlier round. The dung activity term therefore carries 35 %, the width firewood carries.

3.4. Firewood, measured by survey

Firewood is the largest household fuel by energy and the only one with no supply-side statistic. The quantity used here is the HIES 2024–25 design-weighted national total (Pakistan Bureau of Statistics, 2025)—23.1 Mt of air-dry wood a year entering households as purchased, own-produced, in-kind or gifted consumption, recorded in kilograms over a one-month recall for 30,123 households and weighted to the 39.9 million the sampling frame represents.

One vintage mismatch is explicit rather than smoothed: the survey's field year is July 2024 to June 2025 (each household reporting one month within it), while this inventory's reference year is FY2023–24. The round is applied to the adjacent fiscal year because it is the nearest measurement—the alternative is the 2018–19 round, six years out—and the two rounds' 6 percent agreement bounds what a one-year offset can plausibly move; the offset is one of the systematic risks the stream's 35 % uncertainty prices. The time series (Section 7) interpolates per-household quantities between the rounds rather than pinning either to a year it did not measure.

	2018–19	**2024–25**
Reporting households (*n*)	10 636	11 524
Weighted user households (M)	13.0	14.0
kg per user household per day	4.63	4.52
Design SE of the total (%)	2.4[a]	7.1[b]
Top 1% of households' share	0.05	0.19
Top 5% of households' share	0.17	0.33
Interview quarters covered	n/r	4 of 4

Table 2: Firewood survey diagnostics (`hies_diagnostics.csv`). Stratum-clustered linearized SEs: [a]on the published PSU; [b]on an enumeration-block proxy (the 2024–25 release publishes no PSU variable), so the two are not strictly comparable. The 2024–25 round's heavier tail—one percent of reporting households carrying 19 % of the weighted total, against 5 in 2018–19—is the single diagnostic most worth watching; the per-user-day quantities, in the 4.5 kg to 4.6 kg band both rounds, are the reassuring one.

Three estimates now bracket the quantity, and their agreement is the basis for trusting any of them. The independent HIES 2018–19 round (Pakistan Bureau of Statistics, 2019), a different sample six years earlier, gives 22.0 Mt—within 6 %. FAOSTAT's supply series (FAO, 2025), the previous anchor for this stream, sits 7 % above the survey once its oven-dry basis is moisture-matched. A household self-report and a commodity supply balance carry different biases, so their convergence is informative in a way neither alone could be, and the comparison is a published output. FAOSTAT carries FAO's own "E" estimate flag in every year since 2009 and does not move between years, which is why it was demoted to a structural scenario rather than retained as the estimate.

Section 7 still reports the sector three additional ways—excluded, and at ±35 % quantity—so a reader can see exactly what firewood is worth. The 35 % carried in the propagation, down from 60 % in the FAOSTAT era, is not the survey's sampling error—the stratum-clustered design SE is 7.1 % for the 2024–25 round and 2.4 for 2018–19 (Table 2)—but prices what a household survey cannot see: one-month recall scaled to a year, wood consumed outside the dwelling, informal-enterprise use booked to households, and the outlier concentration the diagnostics table quantifies.

3.5. What is deliberately absent

Household electricity. At 39 728 GW h it is the largest single item in every Pakistani household energy statistic—nearly half of national consumption—and it is not in this sector, because those emissions are generated at the power station and counted in 1A1a. Adding them here would double-count roughly a quarter of the national inventory. A test asserts that no electricity stream exists, so a future edit that

"fixes the missing electricity" is rejected rather than silently inflating the country's emissions.

Household coal and charcoal, once NE here, are now the sector's two smallest streams: the HIES processing that re-anchored firewood measured them in passing, at 13.2 kt of coal and 8.3 kt of charcoal—each reported by about one household in a thousand, three orders of magnitude below firewood, and roughly a tenth of what the 2018–19 round recorded. With so few reporting households the survey *suggests* a substantial decline rather than confirms one: the clustered design SEs are 34 % on the 2024–25 coal total (23 reporting households, the heaviest one percent of them carrying 35 % of it) and 16 on charcoal, and questionnaire harmonisation between rounds has not been audited for these two items. The decline direction survives intervals that wide; its magnitude does not. The regulator's nil household coal figure turns out to be roughly right for the wrong reason: not silence, but a quantity too small for a supply table to resolve. One residue caveat is carried in the conservative direction: the crop-residue stream is an *upper-bound residential allocation* of the agriculture-ledger fuel destination. The ledger's fuel fraction covers all fuel use of each residue, not household fuel specifically—brick kilns and small industry burn cotton sticks too—so $R_{\text{fuel}} = R_{\text{household}} + R_{\text{commercial/industrial}} + R_{\text{unresolved}}$, and booking the whole destination to households can only overstate this sector. The survey's price bracket is the demand-side check that bounds it, and the HIES expenditure items are the future constraint on the household share. What remains absent is biogas (digester counts not obtained; NE) and the upstream wood carbonised into the charcoal (roughly 35–50 kt, booked to the producing category IPCC assigns it and unestimated nationally).

4. Methods

4.1. Nine streams, four provenance classes, no household model

For each stream s and gas g,

$$E_{g,s} = A_s \cdot EF_{g,s}, \tag{1}$$

where A_s is the national quantity of fuel entering households in terajoules and $EF_{g,s}$ the IPCC Table 2.5 residential default (IPCC, 2006). The nine streams divide into four provenance classes, and everything else in this section documents how each class produces its A_s: metered-regulatory (natural gas, LPG, kerosene), survey-primary (firewood, dung cake, charcoal, coal), cross-sector ledger (crop residues, and dung cake under configuration L), and expenditure-derived (generators).

With one exception there is no household-count model and no per-capita intensity anywhere in the magnitude chain: a household model would give structure the supply statistics do not, and would introduce a quantity that can drift from the fuel actually supplied. The exception is the reference dung quantity, which is the survey's per-household quantity, interpolated between the two rounds, times the year's households (Section 4.5). Regional and temporal structure enters only downstream of the national totals—the provincial, district, grid and monthly layers distribute each stream's fixed national quantity and reconcile exactly back to it. The transport sector in this inventory makes the same choice for the same reason (Ahmad, 2026).

4.2. Household-survey processing

Both HIES rounds are processed from the PBS SPSS microdata (Pakistan Bureau of Statistics, 2019, 2025): the Section 6A consumption module, item codes 045202 (LPG, kg), 045301 (kerosene, litres), 045401 (firewood, kg), 045402/045403 (charcoal and coal, kg), 045404 (dung cake, kg), and the value-only items 045102 (generator expenses), 045201 (piped gas), 045405/045406 (agricultural wastes and cotton sticks). Each item carries quantity and rupee value over four acquisition routes—purchased, wages-in-kind, own-produced, and gifts or assistance—against a one-month recall; quantities are summed over routes and annualised ×12 (the 2024–25 round's interviews cover all four quarters, so the annualisation averages over season). Design weights are the household weight on the 2024–25 person roster and the published PSU weights for 2018–19; weighted household totals (33.2 and 39.9 million) land on the frame. Own-produced and gifted consumption carries PBS's imputed rupee value, so item values are consumption values, not purchase spending. National totals, province × urban/rural strata, unit values, and the diagnostics of Table 2 (design SEs stratum-clustered on the published PSU or, for 2024–25, an enumeration-block proxy; top-percentile concentration; per-user-day rates) are all committed as derived tables; the microdata itself is fetched and hash-pinned, never redistributed.

The outlier policy is diagnosis before exclusion, and kerosene is the worked example: its 2024–25 item rides 96 reporting households whose implied unit value is a fifth of the retail price, so the survey figure is published as unusable on the comparison table and the stream keeps its OCAC sectoral control. No survey observation is trimmed or winsorised anywhere; concentration is reported (Table 2) rather than removed.

4.3. Mass bases

Section 5.5 (Table 4) states, for every solid fuel, the reported mass and its basis, the moisture and ash fractions, the energy pairing and the pollutant-factor mass. The rule: as-received NCVs pair with air-dry mass, per-kg-dry pollutant factors pair with the dry mass, and no conversion is applied twice.

4.4. The biomass memo convention, and what it does to the arithmetic

Biogenic CO_2 from dung and firewood is reported outside the national total: 12 683 Gg from dung cake, 40 418 Gg from firewood and 12 247 Gg from crop residues and charcoal, 65 347 Gg together (120 805 under configuration L). Their CH_4 and N_2O are counted in full.

The consequence is worth following through, because it produces a result that looks wrong without the convention. Biomass is 65.4 % of household energy and contributes

5.77 Mt of the sector's 24.22 Mt—24 %—entirely as methane and nitrous oxide. Natural gas is 30.5 % of the energy and 66 % of the emissions, because its CO_2 is reportable.

An inventory that reported biomass CO_2 inside the total would show this sector at roughly 90 Mt and would be wrong by IPCC's conventions and by Pakistan's own stated practice (Ministry of Climate Change, Government of Pakistan, 2022). An inventory that excluded biomass CH_4 and N_2O along with its CO_2—a more common error—would show 18.45 Mt and would miss the incomplete-combustion products that are the sector's real climate contribution.

4.5. The dung-cake transfer, and the two configurations

The agriculture sector removes 28 % of cattle and 30 % of buffalo excreted nitrogen from its manure-management balance, because that dung is burned as fuel, and books it IE against Energy. The diverted quantity is that nitrogen as mass:

$$M_{\mathrm{VS}} = \sum_{a} N_a \cdot f_a \cdot \frac{\mathrm{VS}_a}{\mathrm{Nex}_a}, \tag{2}$$

over animal categories a, converted to total solids and air-dry cake through the declared ash and moisture fractions. Configuration L burns all of it in households. The reference (configuration S) instead takes the HIES-reported cake quantity directly—the per-household quantity interpolated between the two rounds and held at the first round's rate before it, times each year's households—and removes the same moisture fraction for its dry mass. The nitrogen equality against agriculture is asserted with == in either case—it concerns what was *diverted*, which neither configuration disputes—and the destination identity is $D_{\mathrm{diverted}} = D_{\mathrm{residential}} + D_{\mathrm{other/unresolved}}$ (Section 7; Appendix C).

4.6. The crop-residue transfer

Every crop's above-ground residue mass closes over four destinations in the agriculture sector's residue ledger,

$$R_{\mathrm{produced}} = R_{\mathrm{returned}} + R_{\mathrm{field\text{-}burned}} + R_{\mathrm{fuel}} + R_{\mathrm{other}}, \tag{3}$$

with the fractions summing to one exactly and $R_{\mathrm{other}} \geq 0$ asserted per crop, so the soils return, the field-burning category and this sector cannot claim the same tonne twice. The residential stream is R_{fuel} (9.27 Mt dry; half cotton sticks)—an upper-bound residential allocation of the ledger's fuel destination. Two caveats are declared: the fuel fraction covers all fuel use, not household fuel specifically ($R_{\mathrm{fuel}} = R_{\mathrm{household}} + R_{\mathrm{commercial/industrial}} + R_{\mathrm{unresolved}}$; conservative for this sector), and its values are Tier 2 judgement—so the survey's price bracket, from the residue expenditure items priced at the survey's own firewood and dung unit values, is published as the demand-side check (7.8 Mt to 24.7 Mt air-dry; the ledger's 10.53 falls inside, asserted by test). The bracket's two ends are priced as named scenarios: 23.92 Mt for the sector at the low end and 25.78 at the high end.

4.7. The generator stream and its road carve-out

Generator fuel is the one expenditure-derived stream: the HIES generator-expenses item (20.6×10^9 rupees, 0.38 % of households) over the administered petrol price (Government of Pakistan, 2025), at standard density, gives 56.3 kt of petrol (2495 TJ), on stationary Table 2.5 factors. Because that petrol is bought at the same stations the road-transport control counts, it is deducted from the road petrol control through the same executable closure the railway diesel carve-out uses (the deduction reads this sector's accessor directly), and a test asserts the equality—the same litre cannot burn in two sectors. Declared limits: the expenditure item may sweep in oil and minor maintenance, the split is assumed all-petrol, and no air-pollutant rows are published (no household-genset factor class exists).

4.8. Air-pollutant factors

Every factor the air layer can burn is a row in the factor file and is rendered, generated rather than transcribed, in Appendix D: Andreae field measurements for the biomass streams (Andreae, 2019), the EMEP/EEA small-combustion chapter for the fossil streams (EMEP/EEA, 2023), agriculture-mirror rows for residues, and black carbon as declared fractions of $PM_{2.5}$. The basis rules (mass versus energy, filterable-condensable caveats, deliberate omissions) are stated there once, and the source-by-pollutant coverage - every E derived from the layer itself, every NE with its declared reason - is the air completeness matrix of Appendix F. All pollutant masses are gross emission at the appliance (Section 8).

4.9. Uncertainty

Parametric uncertainty is IPCC Approach 1 with the activity term of each stream treated as perfectly shared across its gas rows (grouped; the independent treatment is published alongside), and a Monte Carlo counterpart of 10 000 mean-preserving lognormal draws per case ($\sigma = \ln(1 + u/100)/1.96$ from each row's 95 percent half-width; the dung chain shares one correlation group; seed 1404, disjoint from the national inventory's published stream). The dung activity term carries 35 % under the reference, the survey width firewood carries; the ledger chain's 50 % applies to configuration L only. Structural choices are never sampled: the two dung configurations, the construction of the survey dung series (the 2018–19 per-household rate held to the reference year, 23.85 Mt for the sector; the 2024–25 round taken as measured, 24.33), the two firewood anchors, the ends of the residue price bracket and the superseded bases are named cases with conditional intervals and no probability attached. The residential-to-road particulate ratio is sampled with numerator and denominator rows drawn independently, per configuration, and the structural span across configurations is published beside the parametric intervals. The per-input variance decomposition (`residential_variance_decomposition_2024.csv`) attributes the grouped band: firewood factor terms alone carry 67 % of the sector variance.

4.10. Spatial hierarchy

National totals distribute down four levels, reconciling exactly at each: province × urban/rural strata from the HIES quantities; provinces on per-stream geographies (metered gas connections for piped gas; HIES strata quantities for firewood, dung, LPG and residues; population for the four thin-sample streams, with the herd geography retained as dung's structural alternative); districts by the survey's urban/rural rates on census district populations; and a 0.05° grid on per-stream surfaces (built-up fraction for gas and generators, livestock density for dung, cropland for residues, population for the rest—declared judgement of the same class as the railway corridor weights). HIES is representative at province × region, so the within-region district split is population-shaped until the Census 2023 cooking-fuel tables are processed; the gap register carries them.

4.11. Time series and monthly profiles

The FY2015–16 to FY2023–24 series moves gas on a transcribed nine-edition OGRA archive (Oil and Gas Regulatory Authority, 2026) (one interpolated year, labelled), dung and firewood on per-household quantities interpolated between the two survey rounds on census-shaped household counts (configuration L moves dung on the herd through the fixed nitrogen arithmetic), and residues on crop production through the fixed ledger; the five minor streams are held at reference values with basis flags. The reference year is the engine's own output, byte-identical by construction. Below the year, monthly profiles are synthetic and labelled so:

$$E_{m,f} \;=\; (1-h_f)\,\frac{E_f}{12} \;+\; h_f\,E_f\,\frac{\mathrm{HDD}_m}{\sum_{m'}\mathrm{HDD}_{m'}}, \qquad (4)$$

with declared heating fractions h_f per fuel, degree-days from NASA POWER monthly temperature (NASA, 2024) weighted by each province's share of the sector, and the generator stream on the cooling-degree shape instead (load-shedding season). Cooking and other base load is the flat term—the degree-day shape is never applied to it. Twelve months close to every annual total exactly, asserted on every run; the base-versus-climate decomposition is a column in the monthly output.

5. Emission Factors: Table 2.5, Not Table 2.2

5.1. Which of the four tables

IPCC Volume 2 Chapter 2 publishes four parallel default tables for stationary combustion— energy industries, manufacturing, commercial, and residential—and this sector must use Table 2.5, Residential and Agriculture/Forestry/Fishing.

Taking Table 2.2 here would understate household methane by 80 % while leaving CO_2 exactly correct—and CO_2 is 76 % of the sector's reported emissions, so every obvious check would still pass. This is one of four instances of the same trap documented across this inventory (Ahmad, 2026); the power sector's module carries the mirror-image warning, and a test pins the direction in both.

Natural gas	**CO_2**	**CH_4**	**N_2O**
		kg/TJ	
Table 2.5, residential	56 100	5	0.1
Table 2.2, energy industries	56 100	1	0.1
Kerosene			
Table 2.5, residential	71 900	10	0.6
Table 2.2, energy industries	71 900	3	0.6

Table 3: CO_2 is identical because it depends only on the carbon in the fuel. CH_4 is not: residential natural gas is five times the utility value and residential kerosene more than three times, because a small appliance burns less completely than a power-station boiler.

5.2. Biomass factors, and why their bands are wide

The dung and firewood streams carry uncertainties above 150 % on their methane and nitrous oxide. That is not conservatism: cookstove emission factors genuinely span that range, because combustion completeness depends on stove design, fuel moisture, feeding rate and operator behaviour, and measurement campaigns disagree substantially.

The consequence is that the biomass streams that supply most of the sector's energy are also its most uncertain emission streams, which is why the sector's primary band—the grouped Approach-1 figure of ±23.1 % (Section 7)—is wide despite three of the nine streams being metered: the metered gas, LPG and kerosene core is power-station-grade, and everything around it is not.

5.3. Fossil and biogenic methane in one category

This sector is the only one in the inventory that emits both. Dung and firewood methane is biogenic and takes the AR5 GWP of 28; natural gas, LPG and kerosene methane is fossil and takes 30. Every methane-emitting component declares which it is, and a contract test asserts both that the declaration is present and that the CO_2e conversion used the corresponding value.

The difference is small in this sector—methane totals 5.18 Mt CO_2e of which 0.05 is fossil—but the same field distinguishes landfill methane from combustion methane elsewhere in the inventory, and a category emitting both is exactly where a missing flag would be least visible.

5.4. No separate oxidation factor

The 2006 Guidelines fold complete oxidation into the emission factor; Table 1.4 carries an explicit oxidation column of 1.0 for every fuel. Applying the 1996-vintage 0.98 on top would double-discount. The waste sector in this inventory *does* carry an oxidation factor, because open burning genuinely is incomplete, and tests pin both conventions.

5.5. One mass-basis table for every solid fuel

Three mass bases circulate in this sector—as-received (air-dry), total dry solids, and ash-free volatile solids—and the superseded dung estimate showed what conflating them costs. Table 4 states, for every solid fuel, which mass is reported, which mass the energy calculation burns, and which mass the

air-pollutant factors burn, with the moisture and ash fractions that connect them. The rule it encodes: IPCC's Table 1.2 net calorific values are as-received and pair with air-dry mass; Andreae-family pollutant factors are per kilogram of dry fuel including ash and pair with the dry mass; neither conversion is ever applied twice.

6. Validation

6.1. Against EDGAR, at unmatched scope

EDGAR reports Pakistani "buildings" at 31.42 Mt CO_2e for 2024 (Crippa et al., 2025). This inventory reports 24.22 Mt under the reference (77 % of EDGAR) or 29.47 under configuration L (94 %).

The scopes differ in a known direction. EDGAR's buildings category covers residential *plus* commercial and institutional buildings and stationary agriculture; this sector covers households only. A households-only estimate should therefore sit below a buildings estimate, and it does. The remaining 7.20 Mt difference is consistent with EDGAR's broader non-household scope together with methodological and activity-data differences; the direction of the comparison is the check, not its magnitude, and the gap must not be read as a measurement of the commercial sector.

That is a weaker form of validation than a like-for-like ratio, and it is worth being explicit that the comparison cannot detect an error in either direction of comparable size to the scope difference. If this sector were 20 % too low, the ratio would read 64 % and would read as a larger scope difference rather than an error.

6.2. Against Pakistan's own submission

BUR-1 reports 1A4 for FY2017–18 with a residential component of 19.0 Mt, alongside agriculture at 11.3 and commercial at 3.8 (Ministry of Climate Change, Government of Pakistan, 2022). The time series makes the comparison matched-year: this inventory's FY2017–18 value is 22.4 Mt, putting the BUR at 0.85 of it—consistent with a filing that, like the CRT below, carries less than the full biomass scope, though the BUR does not publish its fuel list and the attribution stays a reading, not a finding.

What it does confirm is the convention. Pakistan's own submission states the biomass memo treatment for the same fuels, so this sector's decision to hold 65 347 Gg of biogenic CO_2 outside the total follows national practice as well as IPCC's.

6.3. An executable cross-sector consistency check

The dung nitrogen equality is asserted, not compared: the quantity this sector's ledger receives equals the quantity agriculture's manure balance diverts, by construction and with exact equality, and a test fails if that ever ceases to hold. The equality concerns what was *diverted*—which both configurations accept—not where it was burned; under the reference the residual's destination is an open question, never a conservation failure.

This is a stronger check than any external comparison because it can fail: a change to agriculture's manure-management shares, its herd, or its volatile-solids rates propagates here immediately, where two independently computed dung quantities would drift apart silently and look like two plausible numbers.

6.4. Against the CRT filing, cell by cell

Pakistan's first Common Reporting Tables (filed 2024, inventory year 2022) are extracted from the workbook's own cells on every run, rows found by label, the archive hash pinned by a test. The headline is a notation key: every Biomass cell in the 1.A.4.b block reads NE. The country's two largest household biomass fuels—487 700 TJ in this inventory, 1.6 times the filing's whole residential activity (more than three times under configuration L)—are not estimated in the national submission at all.

Where the filing and this inventory can meet, they agree. The filing's gaseous row carries 288 493 TJ against this inventory's 284 553 from OGRA's domestic table—1.4 % apart, two derivations of one metered quantity that have never seen each other. On the filing's own scope and GWPs at the matched straddle-mean year, the filing reports 17.73 Mt against this inventory's 19.09—within 8 %. The biomass NE is therefore a scope statement, not a magnitude dispute: the two inventories describe the same metered core, and this one adds the fuels the filing leaves out. Two smaller readings off the same block: the liquid IEFs are a proper Table 2.5 LPG/kerosene blend (unlike the road-table substitution the railway block of the same filing carries), and the solid row books 29.5 TJ on the lignite factor set where this inventory carries 341 as other-bituminous—both rounding errors on the sector, recorded rather than smoothed.

6.5. Against the CCAC assessment, as a closing bridge with a residual

The CCAC/SEI integrated assessment (CCAC/SEI, 2023) implies about 224 kt of residential $PM_{2.5}$ for 2021 (35 % of a 641 kt national total). The published bridge steps from this inventory's 283 kt (FY2023–24, reference) to that figure through named, quantified wedges: the inventory year (24 kt, from the time series) and the residue fuel set (68), leaving a signed unexplained residual of −33 kt, −15 percent of the endpoint: PM definition (filterable-versus-condensable scope unresolved on both sides), the GAINS factor family against Andreae field measurements, and remaining activity shares. The bridge closes exactly by construction; the discipline a test enforces is that the residual exists and is never redistributed into the named wedges. The bridge has no dung-configuration wedge: configuration L at the matched year, 591 kt, is a memo row beside it and not a step of it.

Read as evidence on the configuration choice: after the year wedge alone, configuration L stands 2.6 times above the assessment, while the reference lands within 15 % of it. That is not an adjudication—the CCAC estimate is itself a model, its household activity is not a measurement, and agreement with one model is weak evidence—but it is the only independent number that prefers either configuration, and it prefers S.

Fuel	Reported mass	Basis	Moist. (ar)	Ash (dry)	Energy pairing	AP-factor mass
Dung cake (S)	10.93 Mt	air-dry cake, HIES rounds	0.12	0.25	air-dry × 11.6 TJ/Gg (ar)	9.62 Mt total dry solids
Dung cake (L)	58.74 Mt	air-dry cake, from N ledger	0.12	0.25	air-dry × 11.6 TJ/Gg (ar)	51.69 Mt total dry solids
Firewood	23.13 Mt	air-dry wood, HIES 2024–25	0.15	incl. in factors	air-dry × 15.6 TJ/Gg (ar)	19.66 Mt oven-dry
Crop residues	10.53 Mt	air-dry, from residue ledger	0.12	incl. in factors	air-dry × 11.6 TJ/Gg (ar)	9.27 Mt dry matter
Charcoal	8.3 kt	as-purchased, HIES 2024–25	~0 (declared)	incl. in factors	as-purchased × 29.5 TJ/Gg	8.3 kt taken as dry
Coal	13.2 kt	as-received, HIES 2024–25	—	—	as-received × 25.8 TJ/Gg	energy basis (g/GJ)

Table 4: Mass bases, fuel by fuel. The reference (S) reads air-dry cake from the survey, the per-household quantity interpolated between the two rounds, and removes moisture for its dry mass; under configuration L the dung chain runs VS → TS = VS/(1 − ash) → air-dry = TS/(1 − moisture) from the nitrogen ledger. The fractions are the same, so the two configurations differ only in whose quantity they believe, never in chemistry. Charcoal's as-purchased mass is taken as dry (lump charcoal equilibrates at a few percent moisture; the conflation is declared and bounded well below that stream's activity uncertainty). Coal needs no dry mass because its pollutant factors are on fuel energy. Moisture and ash fractions are declared literature-range parameters (Szymajda and Łaska, 2019), pinned to `data/residential_params.csv` by test.

6.6. What is not validated

The within-province geography: HIES is representative at province and urban/rural level, so the district layer's within-region split rides census population, and no district fuel observation tests it. And the air layer's absolute level has no independent Pakistani emissions measurement—the CCAC comparison above is a model-to-model bridge, ambient monitoring exists only as concentrations, and translating those to emitted mass requires dispersion modelling this paper does not attempt.

7. Results

7.1. The sector

Pakistani households emitted 24.22 Mt CO_2e in FY2023–24 on AR5 metrics under the reference (29.47 under configuration L; Table 8), or 24.06 Mt on AR6, plus 65 347 Gg of biogenic CO_2 reported as a memo item outside the national total. This subsection's detail tables ride the reference; every headline is restated under configuration L in Table 8.

Stream	Gas	Gg	Mt CO_2e	±%
Natural gas	CO_2	15 963.42	15.963	10.8
	CH_4	1.42	0.043	150.3
LPG	CO_2	2 142.15	2.142	20.9
Kerosene	CO_2	77.57	0.078	20.2
Coal	CO_2	32.27	0.032	40.3
Generators	CO_2	172.93	0.173	40.2
Dung cake	CH_4	38.05	1.065	154.0
	N_2O	0.51	0.134	154.0
Firewood	CH_4	108.26	3.031	154.0
	N_2O	1.44	0.383	154.0
Crop resid.	CH_4	36.66	1.026	161.6
	N_2O	0.49	0.130	161.6
Total			**24.220**	**22.9**

Table 5: Residential fuel combustion, FY2023–24, AR5. Minor gas rows below 0.01 Mt are omitted from the table and included in the total. Biomass CO_2 is a memo item and does not appear. The total's ±22.9 % combines the rows as if independent; the sector's primary figure is the grouped Approach-1 ±23.1 %, which shares each stream's activity term across its gas rows, and the national Monte Carlo counterpart is ±30.4 %. The last is wider for a named reason: at parameter level the eight biomass CH_4 and N_2O rows' declared widths are superseded by the widths their named factor and calorific-value parameters imply (`data/uncertainty_reconciliation.csv`, the rc41 dispositions), and mean-preserving lognormal draws skew wider still; all are the same decomposition under different treatments, not competing estimates.

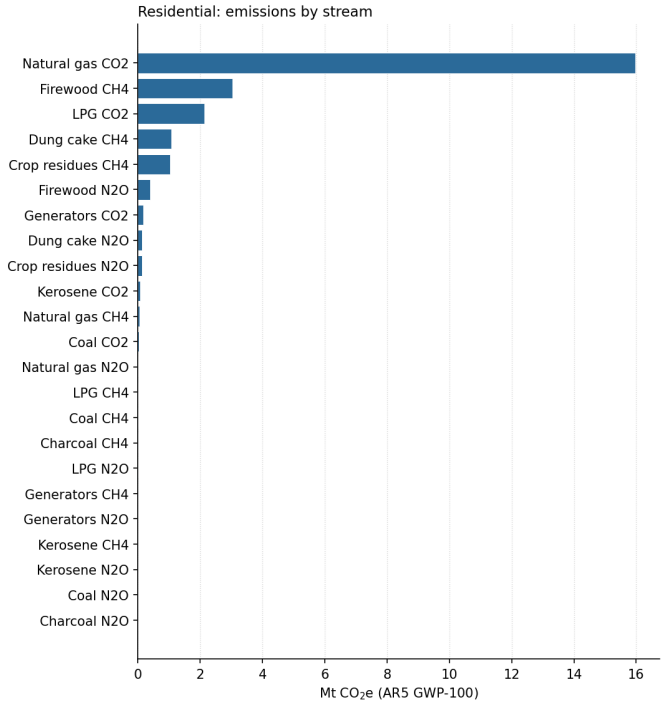


Figure 2: Emissions by stream and gas. Biomass streams appear here through their CH_4 and N_2O only, which is why natural gas carries a far larger share of the emissions than of the energy.

The inversion between energy and emissions is the central arithmetic result. Biomass supplies 65.4 % of household energy and 24 % of reported emissions; natural gas supplies 30.5 % of the energy and 66 % of the emissions. This is the biomass memo convention working as intended, and it is why a household-energy statistic and a household-emissions statistic for Pakistan tell opposite stories about which fuel matters.

7.2. Firewood, and what it is worth

Dropping firewood entirely moves the sector by 14 %—from 24.22 to 20.81 Mt—which is less than a reader might expect for a fuel supplying 39 % of household energy, and the reason is again the memo convention: only firewood's methane and nitrous oxide are in the total. What exclusion *would* remove is 40 418 Gg of biogenic CO_2 from the memo and a comparable share of the air-pollutant layer, where firewood matters most.

Case	Mt CO_2e
Firewood excluded entirely	20.807
Firewood quantity −35%	23.026
Reported	24.220
Firewood quantity +35%	25.415

Table 6: The sector reported four ways across the firewood cases. Firewood is survey-anchored since the HIES re-anchoring; the exclusion case bounds what the stream is worth, and the dung configurations—a larger structural axis—are treated below.

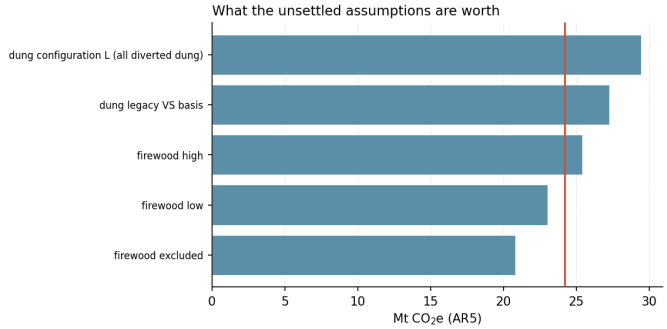


Figure 3: The published sensitivity rows as a range around the reported figure. From lowest to highest: firewood excluded entirely (20.81), the HIES firewood quantity −35% (23.03) and +35% (25.42), the superseded legacy VS dung basis (27.28), which is a reading of configuration L, and configuration L itself (29.47 Mt); the vertical line marks the reference at 24.22 Mt.

7.3. *The survey against the controls*

Fuel	Role	HIES 2024–25	Engine	Ratio
Firewood (Mt)	primary	23.13	23.13	0.94[b]
Dung cake (Mt)[c]	primary	10.93	58.74	0.186
LPG (Mt)	diagnostic	0.90	0.72	1.248
Kerosene (ML)	diagnostic	77.96	30.79	unusable[a]

Table 7: The published survey-versus-control comparison (`residential_hies_comparison_2024.csv`, abridged). Diagnostic ratios are published and never applied. [a]The kerosene item rides 116 outlier-heavy households whose implied unit value is a fifth of retail. [b]Against FAOSTAT's moisture-matched equivalent; the survey IS the estimate. [c]The survey series at the reference year (the per-household quantity interpolated between the two rounds; the 2024–25 round as measured is 11.97 Mt) against the dung agriculture diverts, not against an engine control: the survey IS the estimate, and the ratio is the part of the diverted supply households are observed burning.

The diverted supply is $r_{\text{dung}} = 4.90$ times what households reported in the 2024–25 round, and 5.4 times the survey series at the reference year: the nitrogen ledger implies 58.7 Mt of air-dry cake, and the survey series puts 10.9 in households. Mass conservation constrains what agriculture diverted; it is not evidence about the destination of the conserved mass, and the correct identity is $D_{\text{diverted}} = D_{\text{residential}} + D_{\text{other/unresolved}}$, not $D_{\text{diverted}} = D_{\text{residential}}$. The residential row measures what households burn, and the survey is what observes it; the ledger's burned-fuel shares are unmeasured Tier 2 values, and nothing held supports a fivefold survey under-count. The reference is therefore the survey-observed residential-demand configuration (S), which books the household consumption the survey measures and leaves the residual's destination unresolved, outside the total. Configuration L, in which the full diverted quantity is booked to households, was the reference of earlier releases and is retained beside it as a destination assumption for the diverted dung, not a reading of household use. The survey is not settled either: it sees only in-scope household consumption of formed cake, its two rounds disagree by a factor of 1.53 per household, and in 2024–25 the top 5% of reporting households hold 44.7 % of the reported dung (Section 3).

7.4. *The reference and configuration L*

	L (all diverted)	S (survey, reference)
Dung cake, air-dry (Mt)	58.74	10.93
Household energy (TJ)	1 487 128	932 555
Sector CO_2e, AR5 (Mt)	29.467	24.220
Biogenic memo CO_2 (Gg)	120 805	65 347
Primary $PM_{2.5}$ (kt)	631.9	282.7
Black carbon (kt)	41.5	20.0
Ratio vs road exhaust $PM_{2.5}$	109.4	49.0
National share (%)	6.55	5.44

Table 8: Every headline quantity under both dung configurations (`residential_configurations_2024.csv`). S is the reference, and the detailed spatial and temporal layers are presented under it; under S, 47.8 Mt of diverted air-dry cake has no observed destination and is carried as unresolved, outside the total—candidates include commercial tandoors and eateries, non-household burning, over-stated burned-fuel shares in the manure-management configuration, field application, and survey undercapture of own-produced cake.

The configuration choice is the dominant structural axis of the sector: leaving configuration L for the reference lowers the total by 5.25 Mt, the particulate headline by more than a factor of two, and the road-transport ratio from 109 to 49—all larger than any parametric interval or any other structural case. Section 6 shows the one external anchor that speaks to it: at the matched year, configuration S sits within 15 % of the CCAC/SEI assessment's implied residential particulate, while L is 2.6 times above it.

The 47.8 Mt left unresolved under the reference holds 1025.6 kt of nitrogen and carries no emission. Its candidate destinations are priced one at a time beside the national total, with no probability attached (`national_dung_variants.csv`): burned by non-household users, 5.25 Mt more, which is configuration L reclassified; returned to manure management, 4.49 more; applied to fields, 3.49 more.

7.5. *Nine years and twelve months*

The sector now reports FY2015–16 through FY2023–24: 21.5 to 24.2 Mt, growth carried by gas connections and the survey-anchored biomass streams, with each stream's basis labelled in `residential_timeseries.csv` (gas measured across nine OGRA editions (Oil and Gas Regulatory Authority, 2026) with one labelled interpolation; the minor streams held). The matched calendar years this series unlocks are what Section 6's official comparisons stand on. Below the year, a synthetic-labelled monthly layer on measured degree-days (NASA, 2024) produces a pronounced winter peak that coincides with Pakistan's smog season—January at

49 kt of $PM_{2.5}$ per month against July's 16—with the generator stream alone running the opposite season against summer load-shedding. These are allocations of the annual total, not monthly observations; the output carries the base-load and climate-load components separately, so the degree-day shape is auditable against the flat cooking term it never touches.

7.6. *Provincial allocation*

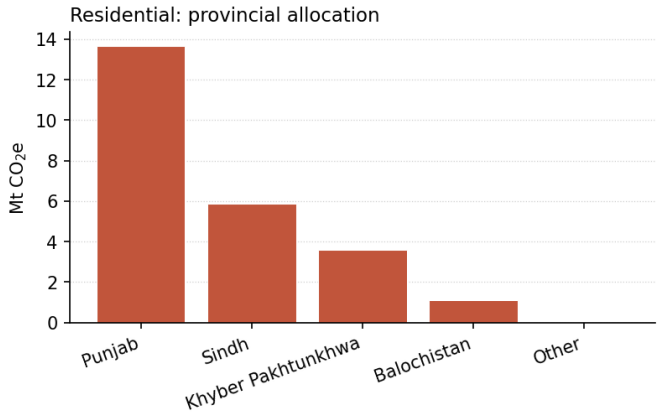


Figure 4: Punjab 13.69 Mt, Sindh 5.86 Mt, Khyber Pakhtunkhwa 3.58 Mt, Balochistan 1.09 Mt. Each stream rides its own proxy; the allocation reconciles exactly against each stream's national figure.

The pattern broadly follows population, as expected for a household sector, with Sindh somewhat elevated relative to its population share because piped gas connection rates in Karachi are high and piped gas is the stream whose CO_2 is reportable.

7.7. *Districts and the grid*

Below the provinces, each stream's total splits urban/rural by the survey's own rates and spreads over 134 census districts by the matching population—the 2023 census's full frame, wider than agriculture's 116 four-province districts because household fuel is surveyed everywhere people live—reconciling exactly per province and stream (`residential_district_2024.csv`); Lahore leads at 2.13 Mt because gas is urban, with Faisalabad (1.01) and Rawalpindi (0.92) behind it. The grid places the same totals at 0.05° on per-stream surfaces, again exactly reconciled (Figures 5 and 6). These are allocations, not observations—no district fuel survey exists—and the within-region split stays population-shaped until the census cooking-fuel tables land (Section 4).

7.8. *Contribution to the national inventory*

Residential fuel combustion is 5.44 % of Pakistan's national emissions and 1.34 % of national Approach-1 variance—a sector whose variance share is below its level share.

It is also almost unaffected by the cross-sector correlated-factor scenario—which, since the common-random-numbers rebuild, moves no sector's band, transport's included: ±30.4 % with parameters shared within the module against ±30.4 %

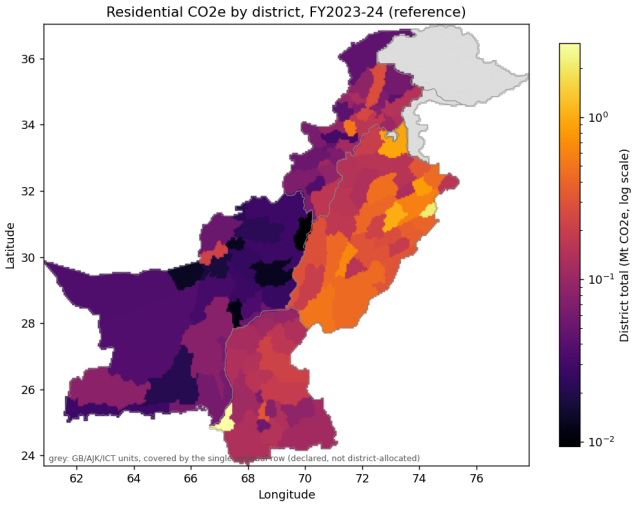


Figure 5: District allocation of residential CO_2e (reference). Each district is painted with its allocated total; the survey's urban/rural rates on census populations do the within-province work. The map now resolves every name through the canonical ADM2 crosswalk: 114 of the 126 boundary polygons are painted (a polygon carrying several census units—Karachi's seven, the Chitral and Kohistan splits—paints their sum), and the 12 grey polygons are the GB/AJK/ICT units covered by the single residual row, a declared class rather than a matching failure; every polygon's disposition is a row of `outputs/residential_district_map_audit_2024.csv`. The map is presentation; the CSV is the product.

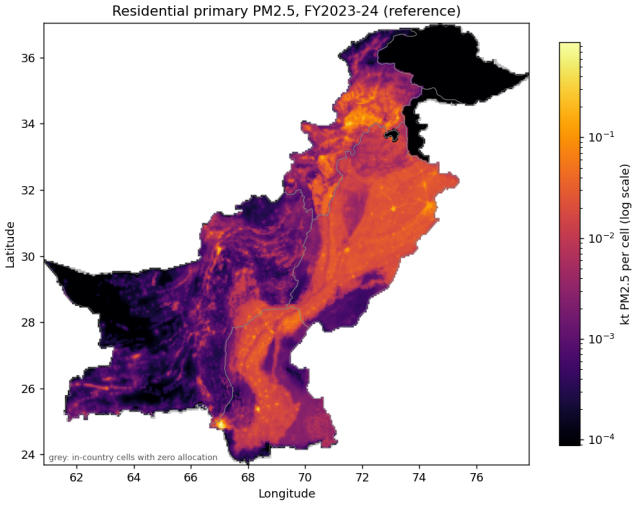


Figure 6: Gridded residential primary $PM_{2.5}$ at 0.05° (reference): gas and generators on built-up fraction (ESA WorldCover v200 2021 class 50, CC BY 4.0, aggregated from 10 m—a 2021 surface against FY2023–24 activity, a declared vintage offset, hashed and documented in `data/gis/PROXY_SOURCES.md`), dung on livestock density, residues on cropland, the rest on population—declared judgement, exactly reconciled to every province and stream total. Colour is logarithmic; light grey marks in-country cells with zero allocation (no proxy weight in that cell), distinct from the white outside the country boundary.

with factors correlated across sectors in the national Monte Carlo (the sampled counterparts of the analytic ±23.1 % grouped figure above). The reason is structural and worth stating as an internal check. That scenario correlates the shared IPCC fuel-carbon factors across sectors, and this sector's uncertainty is dominated by biomass CH_4 and N_2O factors that no other sector shares. A sector that moved materially under it would indicate a factor being shared where it should not be.

8. Air Pollutants: Where This Sector Actually Matters

Household solid fuel is a minor greenhouse-gas sector and Pakistan's dominant source of primary particulate. This section reports the second inventory on the same fuel quantities as the first.

Pollutant	Residential (kt)	Road transport (kt)
CO	3352.8	1430.4
NMVOC	357.6	535.1
$PM_{2.5}$	282.7	8.2
NO_x (as NO_2)	156.1	172.1
Organic carbon	132.0	—
NH_3	42.4	2.6
SO_2	20.8	12.2
Black carbon	20.0	2.0

Table 9: Residential air pollutants (biomass and fossil streams, reference) against road transport's, a sector more than twice its size in CO_2e; configuration L scales the dung rows by 5.37 (Table 8). Both layers derive from their own sector's fuel quantities, so the comparison does not rest on reconciling two activity datasets. NO_x is NO_2-equivalent on both sides; sources publishing as NO are converted on their factor rows. OC and BC are components of $PM_{2.5}$, never additional to it; coal-only PM_{10}/TSP rows live in the factor appendix and the air completeness matrix (Appendix F), not here, because a coal-only coarse fraction beside all-stream $PM_{2.5}$ would mislead.

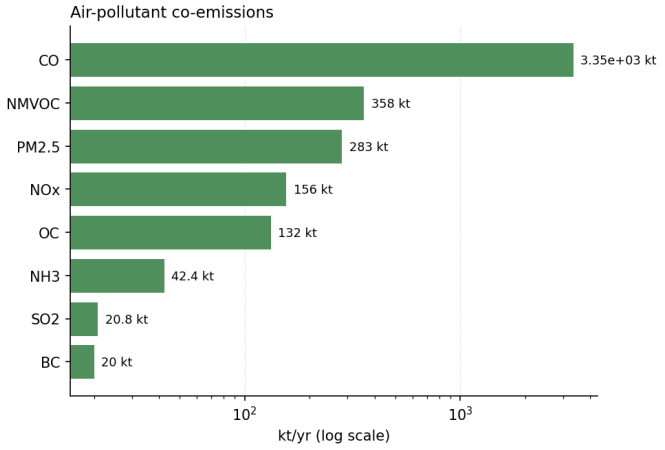


Figure 7: The same figures on a log axis. Carbon monoxide exceeds black carbon by more than two orders of magnitude within one sector.

8.1. The particulate ratio, sampled rather than asserted

Residential fuel burning emits 282.7 kt of primary $PM_{2.5}$ against road transport's exhaust $PM_{2.5}$ of 5.8 kt under the reference—49 times as much (parametric 95 percent interval 14–199)—and 631.9 kt, 109 times (31–445), under configuration L, from a sector less than half road transport's size in CO_2e either way. The interval's width is the honest print of the road layer's EMEP/EEA Tier 1 factor spans, which cover uncontrolled to modern technology for an unmeasured fleet; the superseded per-kilometre diagnostic put the ratio near thirty on factors with no located provenance. Within the national inventory this sector belongs to, household biomass is roughly the largest single source of primary $PM_{2.5}$ under either configuration—roughly two-thirds of the inventory's total under the reference—ahead of open waste burning at 63.9 kt, crop-residue field burning at 54.0 and transport.

The reason this comparison can be asserted rather than hedged is structural. Cross-source particulate comparisons in developing countries usually assemble estimates from separate studies with separate activity bases, so the ratio between two sources is partly an artefact of which datasets each author had. Every air layer in this inventory is computed from its own sector's greenhouse-gas activity data—the same fuel, the same tonnage, the same population panel—which removes that mismatched-activity confounder. What survives in the ratio is everything else: the activity configuration (Section 7), the factor families, and the pollutant and boundary definitions, all stated rather than assumed away. Under matched years and those stated choices, the reference configuration gives the ratio above; it is conditional on them, not a physical constant.

8.2. Black carbon

At 20.0 kt (41.5 under configuration L), household biomass is the dominant Pakistani source of black carbon in this inventory. It is reported as mass and deliberately not converted to CO_2e: its metrics remain contested, its lifetime is days, and a CO_2e figure would imply a comparability with long-lived gases that the science does not support.

That said, the conventional CO_2e total does not capture the sector's short-lived climate-forcing emissions: it is simultaneously the country's largest source of black carbon and of organic carbon, whose forcing effects oppose each other, and no net forcing calculation is presented here. The limitation is that the greenhouse-gas number describes only part of the sector's climate relevance, not that its direction of error is known.

8.3. The policy inversion

The greenhouse-gas and primary-emission rankings of Pakistani sources are close to inverted. A mitigation programme selected on CO_2e would target agriculture, transport and electricity; a programme selected on population exposure would target household cooking and heating first by a wide margin, followed by two open-burning categories that are together 1.0 % of the national inventory.

This is not an argument that greenhouse-gas accounting is the wrong lens. It is an argument that a single inventory cannot rank interventions for both objectives, and that a country choosing air-quality interventions from a greenhouse-gas ranking will systematically under-prioritise the sources that matter most for health.

8.4. Limitations of this layer

The factors are literature values, not Pakistani measurements: Andreae's field compilation for the biomass streams, whose cookstove emissions vary by a factor of several with stove design, fuel moisture and operator behaviour, and the EMEP/EEA small-combustion Tier 1 factors (EMEP/EEA, 2023) for gas, LPG, kerosene and coal—the chapter now held—contributing 16.4 kt of NO_x and under one percent of the biomass $PM_{2.5}$. Generators carry no air rows: no household-genset factor class exists, a declared omission bounded by that stream's 56 kt of fuel. Two factor sensitivities are published in the scenarios table: the technology mix (the reference assumes the traditional/open stove class throughout, consistent with the MTF typology's rural dominance; up to ten percent improved-class adoption would lower $PM_{2.5}$ to 260 kt), and the residue factor family (the crop-residue rows apply field-burning factors as the nearest available class for stove combustion—a provisional assignment; the domestic-biofuel family would give 274).

And the layer reports *gross emitted mass* at the appliance, not mass reaching outdoor air and not exposure. Some indoor particulate deposits before reaching the ambient atmosphere, while personal exposure depends on ventilation, stove placement and behaviour; separating gross emission, outdoor release and exposure requires release fractions and an exposure model this paper does not attempt (the deferred exposure layer in the gap register). The direction of the exposure difference against transport—household emissions released inside and around dwellings, transport emissions dispersing from roadways—is plausible but is not quantified here.

9. Discussion

9.1. One sector, two inventories, opposite rankings

The central result is that this sector is 5.4 % to 6.5 % of Pakistan's greenhouse gases across the two dung configurations while carrying a configuration- dependent but consistently leading share of its primary particulate, and that both readings of the sector are correct.

It is tempting to resolve that tension by arguing that one accounting is wrong—that biomass CO_2 should count, or that particulate should be given a CO_2e equivalent. Neither move is defensible. The biomass memo convention is an accounting rule that prevents double counting: biogenic CO_2 is held outside Energy because any net change in biomass carbon stocks is accounted in AFOLU—whether Pakistan's fuelwood harvest is in fact balanced by regrowth is a land-sector question this paper does not evaluate. Black carbon's metrics genuinely are contested, and converting it would imply a comparability with long-lived gases that the science does not support.

The tension is not an artefact to be resolved but a fact about the sector: it burns a lot of carbon whose net atmospheric effect is booked to the land sector, and emits a lot of particulate that does not warm much but harms a great deal. An inventory that reports only one of those describes half of it.

9.2. What the closure buys, and what it does not

The dung-cake transfer removes a specific failure. Before it, 1.259 Mt of excreted nitrogen a year left agriculture's manure balance correctly and entered nothing—an orphaned flow that both sectors' totals were individually consistent with and that was wrong between them.

What it does not buy is accuracy in the dung quantity itself—or even certainty that households are the destination. The transfer guarantees that the diverted mass is accounted for exactly once, as household combustion or as a typed unresolved remainder; it does not establish that the 28 % and 30 % burned-fuel shares are right, nor that everything they divert is burned in dwellings. The HIES demand-side reading, a factor of 5.4 lower, is why the reference books only what households report burning, why this paper reports the all-diverted configuration beside it (Table 8), and why it restates the identity as $D_{\text{diverted}} = D_{\text{residential}} + D_{\text{other/unresolved}}$. The closure makes the two sectors share one assumption about what was diverted instead of holding two inconsistent ones—an improvement in consistency, while the unresolved remainder carries the open evidence honestly.

That distinction generalises to the other closures in this inventory (Ahmad, 2026): each removes an inconsistency and none removes an unknown.

9.3. The firewood decision

The firewood question used to be whether to include a figure carrying its source's own estimate flag. It is now a different and better question: how far to trust a national household survey.

The answer this sector gives is: as the primary quantity, at 35 %. The HIES total is a design-weighted measurement reproduced within 6 % by an independent round six years earlier, and the supply-side estimate it replaces lands within 7 % of it—three numbers with different biases agreeing at the ten-percent level, which is more corroboration than most activity data in this inventory can claim. The residual 35 % prices the systematic risks a consumption survey cannot see: recall over a one-month window, out-of-dwelling use, enterprise wood booked to households.

The honest reading of the reported figure is a range with two axes, and the configuration axis is the wider one: 24.2 (the reference) to 29.5 Mt (configuration L), not a band around the reference alone. Within the reference, the firewood axis spans 20.8 to 25.4, with the lower bound being the no-firewood case—which is not "the metered fuels alone": it still carries the non-market dung stream, and the metered fossil fuels by themselves support about 18.4 Mt.

9.4. Implications

Household solid fuel is a leading candidate for particulate control. Under the reference and configuration L, residential combustion emits 49 to 109 times the primary $PM_{2.5}$ attributed to road exhaust. This identifies household solid fuel as a leading candidate for particulate-control and exposure assessment—a conclusion that does not follow from the greenhouse-gas inventory and would be missed by a

country prioritising from CO_2e alone—although intervention ranking requires exposure, cost and abatement-effectiveness evidence beyond this inventory.

The two national surveys this sector needed both exist, and both are now held. HIES quantities replaced the FAOSTAT firewood estimate and supply the reference dung quantity—a factor of five below what the nitrogen ledger diverts, with the difference carried as unresolved rather than reconciled away. And the Multi-Tier Framework stove typology (World Bank, 2024) supplies the national stove distribution this paper previously recorded as nonexistent: 44.3 % of households cook mainly on clean-fuel stoves, 71.8 % of rural households on traditional or three-stone/open-fire stoves, with stacking at 6.2 %. The stove-mix sensitivity now prices that evidence; the remaining need is the fuel-by-device-by-end-use crosswalk the report does not publish, which is what separates this inventory from technology resolution.

Fuel switching moves the two inventories in opposite proportions. Replacing biomass with piped gas would *increase* this sector's reported greenhouse gases, because gas CO_2 is reportable and biomass CO_2 is not, while sharply reducing its particulate. Any policy evaluation that scores such a switch on the national inventory alone will score it as a loss. That is a genuine accounting artefact rather than a physical result, and it is the clearest case in this inventory for reporting both layers together.

10. Limitations

Firewood is a self-reported quantity, not a metered one. The HIES total is a real measurement—design-weighted, reproduced by an independent round, corroborated by the supply-side estimate it replaced—but it is households reporting their own consumption over a one-month recall, and the 35 % it carries prices what such an instrument cannot see: recall bias, out-of-dwelling use, enterprise wood booked to households. The sector is still reported with firewood excluded and at ±35 % quantity, and no sensitivity substitutes for a metered series that does not exist.

The dung quantity rests on two survey rounds that disagree. The reference takes household dung from HIES, a per-household quantity interpolated between the 2018–19 and 2024–25 rounds: 10.93 Mt for FY2023–24. The two rounds disagree by a factor of 1.53 per household, and in 2024–25 the top 5% of reporting households hold 44.7 % of the reported dung; the stream carries 35 % for that. The 47.8 Mt of diverted cake the survey does not find in households is unresolved: it sits outside the total with no emission applied, and its candidate destinations are priced beside the total, not inside it. The diverted quantity itself equals what agriculture's nitrogen balance releases, exactly and by construction. That makes the two sectors consistent about what was diverted; it does not establish that the 28 % and 30 % burned-fuel shares are right, and those come from a manure-management configuration that agriculture reports as its own largest uncertainty.

Supply statistics miss non-market fuels. The fossil streams are metered and billed; dung and firewood largely do not pass through a market and are therefore estimated rather than counted. Firewood is the sector's largest energy stream, and the non-market biomass fuels together supply about two-thirds of its energy.

Biomass emission factors span more than a factor of two. The dung and firewood methane and nitrous oxide factors carry uncertainties above 150 %, because cookstove combustion completeness depends on stove design, fuel moisture, feeding rate and operator behaviour, and measurement campaigns disagree. No Pakistani cookstove measurement campaign informs these values.

The fossil air rows are Tier 1 fleet averages with a declared PM ambiguity. Gas, LPG and kerosene pollutants are now estimated on the EMEP/EEA small-combustion chapter's Tier 1 factors (the chapter is held in the repository); the chapter's own review note—that it is unclear whether its PM factors are filterable-only or include condensables—is quoted on the factor rows, and kerosene's European heating-oil SO_2 factor is flagged at ±100 % pending a sulfur-content route.

The layer reports mass, not exposure. Household combustion's health burden depends on proximity, ventilation and time-activity patterns that a national mass inventory cannot capture. The comparison against transport is a source ranking of gross emitted mass, not an exposure ranking; the direction of the exposure difference is plausible but not quantified here (Section 8).

Short-lived forcers are reported and not counted. Black carbon and organic carbon are outside the CO_2e total, deliberately, because their metrics are contested; their forcing effects oppose each other and no net effect is quantified here. The conventional total describes only part of the sector's climate relevance.

Firewood's within-province district and grid proxy is population. The provincial and urban/rural controls come from the survey's own quantities, but below them the within-region split rides census population and the grid rides WorldPop—and firewood availability follows forest and rangeland rather than people, so the within-province pattern is the weakest link in its spatial chain. The national and provincial totals are unaffected.

The built-form bracket has three members and a named absence. Piped gas is allocated on the built-up fraction; the published bracket now prices that choice against 2 admissible rivals—population alone, and the renormalised product of built-up and population—and all three are structurally invariant at district level because the district panel fixes each shape's mass. The choice matters only at the grid cell, where the bracket's spread is published as a per-cell magnitude rather than a district price. The fourth member the ensemble wants—a surface separating residential from non-residential built area—is not held: it is recorded as a named acquisition (EMC-BUILT) rather than approximated by what is on hand.

Mixed vintages inside one sector. The fossil streams are FY2023–24 while firewood rides the 2024–25 survey round and the dung-cake stream a per-household rate interpolated between the two rounds. Stated rather than smoothed over.

The time series is measured where it matters and held

where it does not. The sector now reports FY2015–16 through FY2023–24: gas on a transcribed nine-edition OGRA series (one interpolated year, labelled—the 2020–21 report survives only as an image scan), dung and firewood between the two HIES anchors, residues on crop production. LPG, kerosene and the minor streams—together under four percent—are held at their reference values with basis flags; their archive series is recorded as remaining work, and per-household firewood and dung before the 2018–19 round are an assumption, not a trend claim.

11. Conclusion

Pakistani households emitted 24.2 Mt to 29.5 Mt CO_2e in FY2023–24 across the survey-based reference and the all-diverted dung configuration—where the diverted dung that households do not report burning goes is the sector's dominant open question—plus biogenic CO_2 reported outside the total. The same combustion emitted 283 kt to 632 kt of gross primary $PM_{2.5}$: under either configuration the sector is among the country's largest primary-particulate sources, at 49 to 109 times road transport's exhaust particulate, from a sector less than half road transport's size in CO_2e.

Three results generalise. First, one sector can be minor in a climate inventory and dominant in a primary-emission inventory, and both readings can be right: the biomass memo convention holds biogenic CO_2 outside national totals while counting the methane and nitrous oxide of incomplete combustion, so a country selecting air-quality interventions from a greenhouse-gas ranking will systematically under-prioritise the sources that matter most for health. Second, computing both emission layers from one activity basis is what makes their comparison meaningful: under matched years and the stated definitions, the particulate ratio reflects combustion and fuel quantities rather than which datasets two authors happened to hold. Third, a cross-sector transfer should be a derivation rather than a second estimate—and a conservation constraint should never be mistaken for destination evidence: the ledger fixes what agriculture diverted, the survey observes what households burned, and the factor-of-five gap between them is reported as the open question it is.

Both national surveys this sector needed turned out to exist. HIES replaced the flagged FAOSTAT firewood figure with a twice-corroborated measurement and exposed the dung disagreement; the Multi-Tier Framework stove typology (World Bank, 2024) feeds the technology-mix sensitivity. What remains for full resolution is the fuel-by-device-by-end-use crosswalk, Pakistani emission measurements, and the exposure layer this paper's scope deliberately excludes.

Code and data availability

The calculation engine, the activity data and every output quoted here are version-controlled in the inventory repository (https://github.com/RehanAhmad13/pakistan-emissions-sectors) and reproduce from a single command. The sector is regenerated with `python3 -m pkinv.sectors.residential.run_residential`; Appendix A gives the full sequence. Every number printed in this paper is read from a generated CSV rather than transcribed, and a test asserts that the manuscript and the engine agree.

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Appendix A. Reproduction

The sector runs on Python 3.11 with no compiled dependencies. From the repository root:

```
pip install -e .
python3 -m pkinv.sectors.residential.run_residential
python3 -m unittest discover tests
```

The first command runs every layer of the sector in sequence—the greenhouse-gas engine, the air-pollutant layer, the survey comparison, the official benchmarks and CCAC bridge, the uncertainty module, the district allocation, the time series and the monthly profiles—and prints the dung-cake nitrogen closure as it runs. Its outputs land in `outputs/`, and this is which is which—the list they replace named them collectively as `residential_*.csv`, which told a reader what exists without giving a filename for any of it:

- `residential_totals_2024.csv`: the sector total under the reference, with configuration L beside it and the sector's mixed input vintages stated on the row rather than smoothed over.
- `residential_by_stream_2024.csv`: the same total by fuel stream.
- `residential_fuel_2024.csv`: the energy behind those streams in TJ and share.
- `residential_provincial_2024.csv`: the provincial split, and `residential_district_2024.csv` the district allocation beneath it.
- `residential_strata_2024.csv`: the allocation basis: fuel by province and rural or urban stratum, on the survey's own strata.
- `residential_monthly_2024.csv`: the monthly profile, and `residential_emissions_monthly_2024.csv` the emissions it carries.
- `residential_air_pollutants_2024.csv`: the air-pollutant layer, and `residential_pm_ratio_2024.csv` the residential-over-road primary $PM_{2.5}$ ratio with its parametric interval.
- `residential_hies_comparison_2024.csv`: the survey comparison, this inventory's quantities against the HIES round that anchors them.

- `residential_official_comparison_2024.csv`: the official benchmarks, and `residential_unfccc_crt_cell_audit_2024.csv` the CRT cell extraction behind them.
- `residential_ccac_bridge_2024.csv`: the CCAC bridge, item by item in $PM_{2.5}$ terms.
- `residential_uncertainty_2024.csv`: the interval; `residential_uncertainty_inputs_2024.csv` the input register behind it; `residential_uncertainty_scenarios_2024.csv` the structural scenarios.
- `residential_dung_ledger_2024.csv`: the dung-cake closure the run prints: each claimant's air-dry mass, its share of the diverted quantity and how it is treated.
- `residential_timeseries.csv`: the series across years.
- `residential_qa_2024.csv`: the quality-control report.

Figures regenerate with `pkinv.report.figures` and are copied into this manuscript by `scripts/sync_figures.py`.

The dung stream's diverted quantity is derived from the agriculture sector's nitrogen flow, so reproducing it requires that sector's data to be loadable; the equality of the two sectors' nitrogen is asserted by `tests/test_residential.py` with == rather than a tolerance, and the destination configurations (L/S) share that quantity.

The HIES microdata are not redistributed here: the SPSS files are fetched from the Pakistan Bureau of Statistics microdata portal and hash-pinned in a manifest, and only the derived, design-weighted aggregate tables are committed. Every other input is committed directly, and a run manifest hashes the held sources.

Appendix B. Sources

Natural gas and LPG supplied to households are OGRA's sector-wise tables (Oil and Gas Regulatory Authority, 2024), with the nine-edition archive (Oil and Gas Regulatory Authority, 2026) behind the time series; household kerosene is OCAC's sectoral sales (Oil Companies Advisory Council, 2024). Firewood is the HIES 2024–25 design-weighted household quantity (Pakistan Bureau of Statistics, 2025), with the 2018–19 round (Pakistan Bureau of Statistics, 2019) as an independent check and FAOSTAT's wood-fuel series (FAO, 2025) retained only as a structural scenario, carrying FAO's own estimate flag. Household dung cake is the HIES quantity, a per-household rate interpolated between the two rounds (Pakistan Bureau of Statistics, 2019, 2025); the diverted quantity it is set against, and configuration L, are derived from the agriculture sector's nitrogen balance rather than from any external source, with cake ash and moisture from proximate-analysis literature (Szymajda and Łaska, 2019). Greenhouse-gas factors are IPCC 2006 Guidelines Volume 2 Chapter 2 Table 2.5 (IPCC, 2006); air-pollutant factors are the Andreae compilation (Andreae, 2019), the EMEP/EEA small-combustion chapter (EMEP/EEA, 2023), the Akagi compilation (Akagi et al., 2011), and the field-burning mirrors (EMEP/EEA, 2019) for crop residues. Degree-days derive from NASA POWER monthly temperature (NASA, 2024); the generator stream uses the Government of Pakistan's administered petrol prices (Government of Pakistan, 2025). External comparisons are EDGAR (Crippa et al., 2025), Pakistan's CRT workbook (Government of Pakistan, 2026), BUR-1 (Ministry of Climate Change, Government of Pakistan, 2022) and the CCAC/SEI assessment (CCAC/SEI, 2023).

Appendix C. Supplementary Note: the dung-cake closure

The design decision worth recording is what the ledger conserves and what it leaves open. It conserves the DIVERTED quantity: agriculture removes 28 % of cattle and 30 % of buffalo excreted nitrogen from its manure-management balance and books it IE to Energy, and this sector receives exactly that quantity—not an independent estimate that would then need ongoing reconciliation, whose failure mode is silent (both sectors' totals stay internally consistent while the flow between them does not balance). What the ledger does NOT settle is the destination of the diverted mass. The household survey observes 11.97 Mt of cake consumed in dwellings in its 2024–25 round, a factor of 4.9 below the diverted total, and the series interpolated between its two rounds puts 10.93 there in FY2023–24. The paper reads the destination two ways: the reference books the observed household consumption and carries the remainder as $D_{\text{other/unresolved}}$, outside the total with no emission applied, and configuration L, the reference of earlier releases, books it all to household combustion (Section 7).

The diverted quantity itself is derived from agriculture's nitrogen flow:

$$M_{\text{VS}} = \sum_a N_a \, f_a \, \frac{\text{VS}_a}{\text{Nex}_a}, \tag{C.1}$$

using the same herd, the same species categories, the same burned-fuel shares f_a, and IPCC's volatile-solids and nitrogen-excretion rates for the same animal categories and region. The result is 1.259 Mt of nitrogen corresponding to 38.77 Mt of ash-free volatile solids—the ledger intermediate, not a burnable fuel mass—and a test asserts the correspondence with == rather than a tolerance.

Three masses follow from that one intermediate, and each pairs with its own factor basis: total dry solids restore the mineral ash a hand-formed cake carries (TS = VS/(1 − ash), 51.69 Mt) and are the basis Andreae's per-kilogram-dry-fuel pollutant factors describe; air-dry cake mass restores the equilibrium moisture (TS/(1 − moisture), 58.74 Mt) and pairs with IPCC's as-received net calorific value for the energy calculation. These are the masses of the diverted supply, and of the stream configuration L burns; the reference reads air-dry cake from the survey and removes the same moisture fraction for its dry mass (Table 4). The superseded pairing—ash-free volatile solids on the as-received NCV, a three-bases conflation

that understated the stream—is retained only as the structural scenario `dung_basis_legacy_vs`.

The property this buys is not accuracy but the impossibility of a particular error. If agriculture's manure-management shares change, the diverted quantity changes with them, in the same commit, and neither sector can be updated into inconsistency with the other about what was diverted. Where the part of that quantity households do not report burning goes remains the sector's dominant open question, carried explicitly rather than resolved by construction.

Appendix D. Supplementary Table: every air-pollutant factor the layer burns

Basis rules, stated once: biomass factors are per kilogram of dry fuel including ash (total dry solids for dung, oven-dry for firewood, dry matter for residues, as-purchased for charcoal); fossil factors are per gigajoule of fuel energy; black carbon is reported INSIDE $PM_{2.5}$, never additional to it; NO_x is reported as NO_2-equivalent throughout—sources publishing as NO (the Andreae rows) are converted $\times 46/30$ on their data rows, so one basis sums; the device is a single implicit class per fuel—the traditional/open class that dominates rural biomass cooking (71.8% of rural households per the MTF 2021–22 stove typology; the fuel-by-device crosswalk needed for full technology resolution is not yet published). All values are gross emission at the appliance. Flags: c = the source's own filterable-versus-condensable caveat is quoted on the data row; d = the printed confidence interval is defective and the row carries ± 100 with the bounds quoted; s = a sulfur-content assumption is stated on the row.

Pollutant	Fuel	Value	Unit	Source	Flags	±%
$PM_{2.5}$	Charcoal	18.7	g/kg dry	Andreae 2019	–	85
$PM_{2.5}$	Coal	398	g/GJ	EMEP/EEA 2023 ch. 1.A.4	c	51
$PM_{2.5}$	Crop residues	7.6	g/kg dry	Andreae 2019	–	50
$PM_{2.5}$	Dung cake	8.3	g/kg dry	Andreae 2019	–	40
$PM_{2.5}$	Firewood	6.7	g/kg dry	Andreae 2019	–	49
$PM_{2.5}$	Kerosene	1.9	g/GJ	EMEP/EEA 2023 ch. 1.A.4	c	39
$PM_{2.5}$	LPG	1.2	g/GJ	EMEP/EEA 2023 ch. 1.A.4	c	42
$PM_{2.5}$	Natural gas	1.2	g/GJ	EMEP/EEA 2023 ch. 1.A.4	c	42
PM_{10}	Coal	404	g/GJ	EMEP/EEA 2023 ch. 1.A.4	c	50
TSP	Coal	444	g/GJ	EMEP/EEA 2023 ch. 1.A.4	c	59
BC	Charcoal	0.43	g/kg dry	Andreae 2019	–	49
BC	Crop residues	0.5	g/kg dry	Akagi 2011	–	60
BC	Dung cake	0.51	g/kg dry	Andreae 2019	–	67
BC	Firewood	0.53	g/kg dry	Andreae 2019	–	66
BC	Gaseous fuels	0.054	frac. of $PM_{2.5}$	EMEP/EEA 2023 ch. 1.A.4	–	77
BC	Liquid fuels	0.085	frac. of $PM_{2.5}$	EMEP/EEA 2023 ch. 1.A.4	–	72
BC	Solid fuels	0.064	frac. of $PM_{2.5}$	EMEP/EEA 2023 ch. 1.A.4	d	100
OC	Charcoal	5.9	g/kg dry	Andreae 2019	–	42
OC	Crop residues	3.3	g/kg dry	Akagi 2011	–	60
OC	Dung cake	4.4	g/kg dry	Andreae 2019	–	43
OC	Firewood	3.0	g/kg dry	Andreae 2019	–	50
CO	Charcoal	207	g/kg dry	Andreae 2019	–	30
CO	Coal	4600	g/GJ	EMEP/EEA 2023 ch. 1.A.4	–	43
CO	Crop residues	92	g/kg dry	IPCC 2006 V4 (mirror)	–	40
CO	Dung cake	89	g/kg dry	Andreae 2019	–	47
CO	Firewood	83	g/kg dry	Andreae 2019	–	35
CO	Kerosene	57	g/GJ	EMEP/EEA 2023 ch. 1.A.4	–	40
CO	LPG	26	g/GJ	EMEP/EEA 2023 ch. 1.A.4	–	46
CO	Natural gas	26	g/GJ	EMEP/EEA 2023 ch. 1.A.4	–	46
NO_x	Charcoal	1.81	g/kg dry	Andreae 2019	–	73
NO_x	Coal	110	g/GJ	EMEP/EEA 2023 ch. 1.A.4	–	75
NO_x	Crop residues	2.5	g/kg dry	IPCC 2006 V4 (mirror)	–	50
NO_x	Dung cake	4.29	g/kg dry	Andreae 2019	–	46
NO_x	Firewood	3.83	g/kg dry	Andreae 2019	–	52
NO_x	Kerosene	51	g/GJ	EMEP/EEA 2023 ch. 1.A.4	–	40
NO_x	LPG	51	g/GJ	EMEP/EEA 2023 ch. 1.A.4	–	39
NO_x	Natural gas	51	g/GJ	EMEP/EEA 2023 ch. 1.A.4	–	39
NMVOC	Charcoal	6.6	g/kg dry	Andreae 2019	–	74
NMVOC	Coal	484	g/GJ	EMEP/EEA 2023 ch. 1.A.4	–	61
NMVOC	Crop residues	7.0	g/kg dry	Akagi 2011	–	60
NMVOC	Dung cake	14.4	g/kg dry	Andreae 2019	–	70
NMVOC	Firewood	7.8	g/kg dry	Andreae 2019	–	64
NMVOC	Kerosene	0.69	g/GJ	EMEP/EEA 2023 ch. 1.A.4	–	43
NMVOC	LPG	1.9	g/GJ	EMEP/EEA 2023 ch. 1.A.4	–	39
NMVOC	Natural gas	1.9	g/GJ	EMEP/EEA 2023 ch. 1.A.4	–	39
SO_2	Coal	900	g/GJ	EMEP/EEA 2023 ch. 1.A.4	s	39
SO_2	Crop residues	0.40	g/kg dry	EMEP/EEA 2019 (mirror)	–	70
SO_2	Dung cake	0.77	g/kg dry	Andreae 2019	–	48
SO_2	Firewood	0.47	g/kg dry	Andreae 2019	–	94
SO_2	Kerosene	70	g/GJ	EMEP/EEA 2023 ch. 1.A.4	d,s	100
SO_2	LPG	0.3	g/GJ	EMEP/EEA 2023 ch. 1.A.4	–	33
SO_2	Natural gas	0.3	g/GJ	EMEP/EEA 2023 ch. 1.A.4	–	33
NH_3	Charcoal	2.5	g/kg dry	Andreae 2019	–	70
NH_3	Coal	0.3	g/GJ	EMEP/EEA 2023 ch. 1.A.4	d	100
NH_3	Crop residues	1.3	g/kg dry	Akagi 2011	–	60
NH_3	Dung cake	1.33	g/kg dry	Andreae 2019	–	59
NH_3	Firewood	0.89	g/kg dry	Andreae 2019	–	55

Table 10: The complete air-pollutant factor set, generated from the factor file the engine reads (`data/emission_factors.csv`). Omissions are deliberate and declared in the text: charcoal SO_2 (unusable source cell), NH_3 for gaseous and liquid fuels (not applicable in the chapter's tables), and every generator pollutant (no household-genset factor class exists).

Appendix E. Supplementary Table: the 1.A.4.b completeness matrix

Every category–gas combination the sector might contain, with its IPCC notation key and the justification the engine carries for every non-E entry. Generated from the same completeness table the national inventory publishes, which is itself synced to the gap register by test.

Category	Key	Gases	Justification (abridged)
piped natural gas	E	CO2/CH4/N2O	OGRA sector-wise gas consumption (FY2023-24, 821 MMCFD) x IPCC Vol 2 Ch 2 Table 2.5, the Residential table - not the Energy Industries table used by 1.A.1.a.
LPG	E	CO2/CH4/N2O	OGRA national supply x a domestic share read from OGRA Figure 5.2, which is published only as a bar chart.
kerosene	E	CO2/CH4/N2O	OCAC total SKO, treated in full as household use; an upper bound, but under 0.2 Mt CO2e either way.
dung cake	E	CH4/N2O	The household survey's quantity (HIES 2018-19 and 2024-25, per-household rate interpolated). The IE transfer above is the diverted supply it is set against; the part with no household destination is unresolved and outside the total. Biogen…
firewood	E	CH4/N2O	HIES 2024-25 design-weighted household quantity: 23.1 Mt air-dry wood, a measurement replacing FAOSTAT's flagged, unmoving estimate. Twice corroborated - HIES 2018-19 agrees within 6%, FAOSTAT's moisture-matched equivalent within 7% - an…
biogenic CO2 (dung cake, firewood, residues, charcoal)	IE	CO2	Memo item, outside the national total, per IPCC's biomass convention and Pakistan's own BUR-1. Any net stock loss belongs in LULUCF.
electricity	IE	-	Household electricity is nearly half of national consumption and is counted at the point of generation in 1.A.1.a. Deliberately absent; a test asserts no electricity stream exists.
coal and charcoal	E	CO2/CH4/N2O	HIES 2024-25 design-weighted quantities - 13.2 kt coal, 8.3 kt charcoal, each on about one household in a thousand, three orders below firewood. OGRA's nil household coal row was roughly right for the wrong reason (a quantity too small for…
crop residues and cotton sticks	E	CH4/N2O	An upper-bound residential allocation of the agriculture residue ledger's fuel destination: frac_fuel covers all fuel use (R_fuel = R_household + R_commercial/industrial + R_unresolved), and the whole 9.3 Mt dry (10.5 Mt air-dry) is b…
petrol generators	E	CO2/CH4/N2O	HIES 2024-25 generator-fuel expenditure over the administered petrol price: 56 kt of petrol, 2,495 TJ, 0.17 Mt CO2e. Carved out of the road-transport petrol control through the same relayed-accessor idiom as railway diesel, asserted with e…
biogas digesters	NE	CH4	Domestic biogas digesters exist in Pakistan (PCRET-supported units, cattle-belt provinces); the programme installation records that would count them have not been obtained here, and HIES carries no biogas consumption item to fall back on.…
charcoal production wood	NE	CO2/CH4	The 8.3 kt of charcoal burned implies roughly 35-50 kt of wood carbonised upstream (4-6 kg wood per kg charcoal), with kiln CH4 that IPCC books to the producing category. No Pakistani charcoal-production statistic exists; the quantity is b…

Table 11: The residential completeness matrix, generated from the engine (`outputs/completeness.csv`, 1.A.4.b rows). Full justifications, including the magnitudes that bound each NE entry where a comparator exists, are in the generated national table.

Appendix F. Supplementary Table: the air-pollutant completeness matrix

Every stream–pollutant combination, E derived from the air layer's actual rows (the matrix cannot claim coverage the engine does not compute) and every NE carrying its declared reason, keyed below the table. All estimated cells are gross emitted mass at the appliance, on the reference (household-survey) quantity for the dung row.

Stream	$PM_{2.5}$	PM_{10}	TSP	BC	OC	CO	NO_x[n]	NMVOC	SO_2	NH_3
Natural gas	E	NE[a]	NE[b]	E	NE[c]	E	E	E	E	NE[d]
LPG	E	NE[a]	NE[b]	E	NE[c]	E	E	E	E	NE[e]
Kerosene	E	NE[a]	NE[b]	E	NE[c]	E	E	E	E	NE[f]
Coal	E	E	E	E	NE[c]	E	E	E	E	E
Generators	NE[g]	NE[g]	NE[g]	NE[g]	NE[g]	NE[g]	NE[g]	NE[g]	NE[g]	NE[g]
Dung cake	E	NE[a]	NE[b]	E	E	E	E	E	E	E
Firewood	E	NE[a]	NE[b]	E	E	E	E	E	E	E
Charcoal	E	NE[a]	NE[b]	E	E	E	E	E	NE[h]	E
Crop residues	E	NE[a]	NE[b]	E	E	E	E	E	E	E
Biogas	NE[i]	NE[i]	NE[i]	NE[i]	NE[i]	NE[i]	NE[i]	NE[i]	NE[i]	NE[i]

Table 12: The air-pollutant completeness matrix. [n]NO_2-equivalent basis throughout. NE reasons: [a] Andreae tabulates TPM, not PM10; the coarse fraction is not adopted for biomass and the gaseous/liquid chapters' PM10 adds nothing above PM2.5 for premixed flames. [b] as PM10: not adopted for biomass; the coal row is the one solid fuel with a chapter TSP factor. [c] no OC factor adopted for this stream. [d] not applicable in the chapter's gaseous-fuel table; omitted, never zero-filled. [e] not applicable in the chapter's gaseous-fuel table. [f] not applicable in the chapter's liquid-fuel table. [g] no household-genset factor class exists in the held EMEP chapter; declared omission bounded by the stream's 56 kt of fuel (the non-road small-engine chapter is the registered upgrade). [h] the source cell prints a range its own average falls outside, on two measurements - unusable, omitted rather than guessed. [i] the stream itself is NE (no digester count exists; see the completeness matrix and residential-biogas in the gap register).